

DEVIN TAKE-HOME · COGNITION

Autonomous Dependency Security

Turning never-ending CVE toil into a hands-free remediation loop - with Devin doing the work.

Presented by **Deon Menezes** · live on a fork of Apache Superset

Dependency security is endless toil for engineering teams

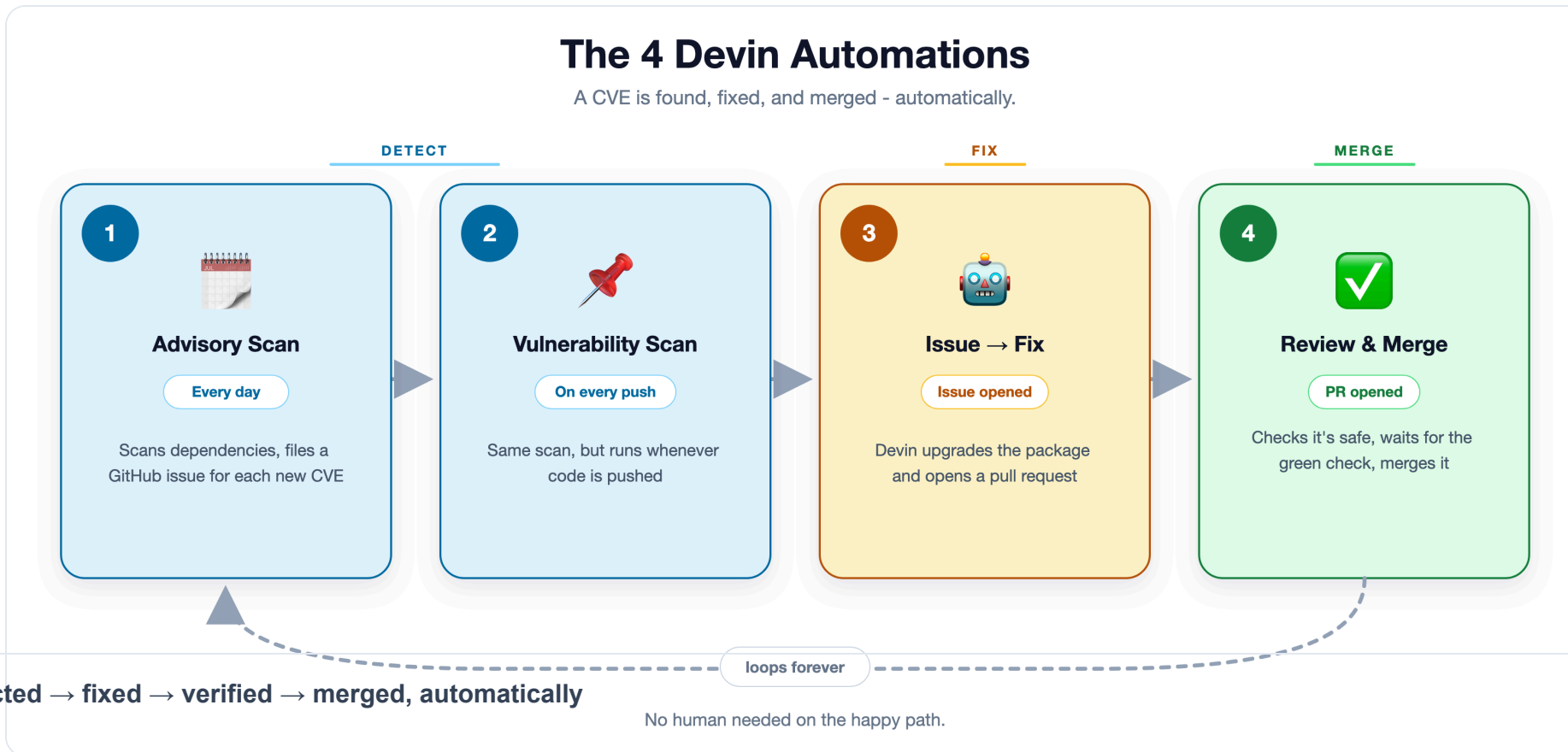
- ◆ Scanners file a steady **drip of CVE issues** - every week, forever
- ◆ Each one is the same grind: **triage** → **bump** → **fix conflicts** → **test** → **PR** → **merge**
- ◆ Low-value, high-interrupt work that pulls **senior engineers off the roadmap**
- ◆ Skip it and **security debt and audit risk compound**

Important, tedious, and never actually done - the perfect job to automate.

An event-driven loop where Devin does the fixing

The 4 Devin Automations

A CVE is found, fixed, and merged - automatically.



The loop in action

- ◆ A scan files a **devin-remediate** issue for each new CVE
- ◆ Devin opens a scoped PR - “**security: upgrade X → Y**”, “**Closes #N**”
- ◆ The engine posts the **deps-verify** check; branch protection requires it
- ◆ Devin auto-reviews scope + green check, then **squash-merges** and the issue closes
- ◆ A live **control room** shows every run, PR link, and ACU spent

**One CVE issue in → one merged, verified pull request out.
No human on the happy path.**

HOW · KEY DECISIONS IN THE CODE

- ◆ **Group by package** - one PR per package, not one per CVE
- ◆ **Guardrails**: no published fix, or a **major-version bump** → held for a human
- ◆ **The engine is the CI** - posts a first-class **deps-verify** status (private fork, Actions off)
- ◆ **Dry-run, a run ledger, and a report** so a lead can see it working

HOW · THE FULL LOOP

Every path - including the guardrails

Green = automated · amber = decision · red = human hand-off

This needs an agent, not a script

- ◆ A script bumps a pin. **Devin reads the repo**, regenerates lockfiles, resolves conflicts, fixes compatibility, runs tests, and writes the PR
- ◆ Handles the **long tail** - transitive pins and code changes - that break naive bots
- ◆ Native **automations trigger on GitHub events** - no glue infrastructure to run
- ◆ **Judgment**: it recognizes when a fix isn't safe and stops

The value isn't the version bump. It's everything around the bump.

PROOF · IT WORKS, END TO END

Live on a real repository

15 / 17

CVEs remediated and merged - hands-free

2 held

correctly: flask (major migration) & paramiko (no fix)

0

un-verified merges - every PR gated by deps-verify

4

Devin automations, chained into one closed loop

How this scales past the demo

- ◆ Make the gate **semantic** - run the real test suite, not just pin integrity
- ◆ **Multi-repo, multi-ecosystem** - npm, cargo, go.mod via per-ecosystem parsers
- ◆ Notify **Slack / PagerDuty**; severity-aware **SLAs** for critical CVEs
- ◆ **Per-team policy**: auto-merge patches, review minors, escalate majors
- ◆ One source of truth - **Postgres + a metrics backend** as volume grows

THANK YOU

Dependency security, on autopilot

Detect → fix → verify → merge. Humans only
on the exceptions.

Deon Menezes · built with Devin